Modeling Basketball Production with a Zero-Inflated Poisson Distribution

DR. OLGA STAT 410

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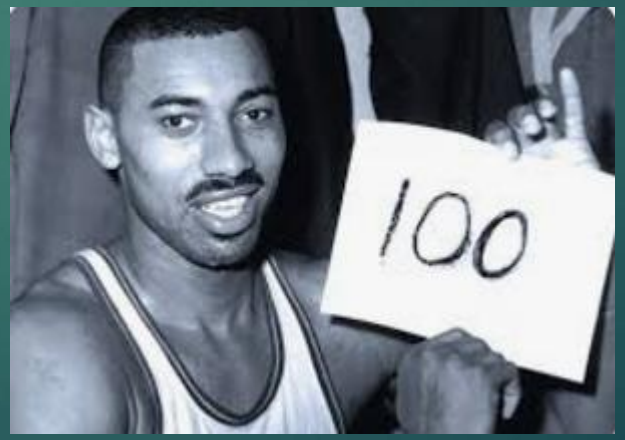


Why this topic?

- The NBA is both fun to watch and produces a plethora of individual and team-based statistics each game.
- Which statistics are “empty calories”, and which contribute to positive team performance?
- Advanced stats like PER (player efficiency rating) and WAR (wins above replacement) are the first steps in quantifying a player’s “worth”, but there are better ways.

ZIP Model Premise

SAS and Zip Bayless

- The presence of structural zeros when looking at points scored in a game by individual players makes a Zero-Inflated Poisson model optimal
- The structural zeros of points scored come from players not playing significant minutes (from injury or strategic team matchups).

Data Set

GAME_ID	Position	Pts	Min	PLAYER_NAME	TeamCity	PlusMinus	Height	YearsLeague
22200477	F	2	18.06	Romeo Langford	San Antonio	-2	6.42	3

- Kaggle dataset of all NBA games played from 2004 to winter 2022, limited to the 200 most recent player performance observations, 8 regular season games on December 21st and 22nd, 2022.

- The player position variable has some missingness to manually fill (usually Wikipedia), and data was added for player height and years played.
- Variables of interest are Points Scored, Starting Position or Position Played, Minutes Played, Team, Plus/Minus, Height, and Years in the League (as of 2022).

Notable numerical variables summarized

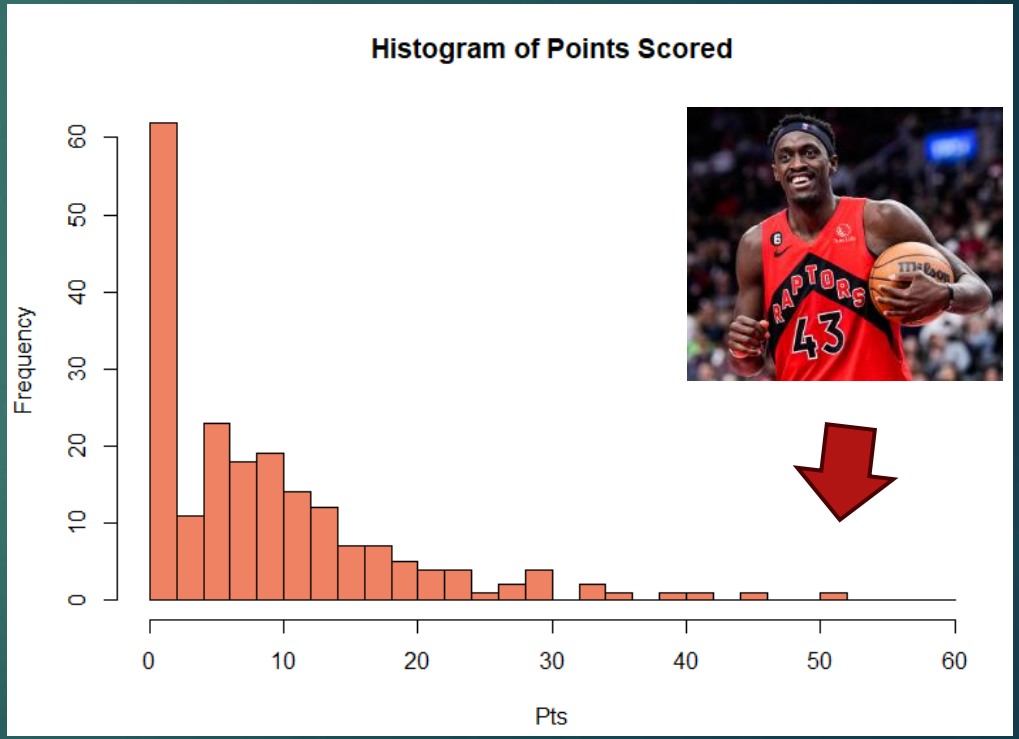
```
hist(nba.data$Pts,
     breaks = seq(0, 60 , by = 2),
     main = "Histogram of Points Scored",
     xlab = "Pts",
     ylab = "Frequency",
     col = "salmon2",
     border = "black")
```

```
nba.data<- read.csv(file="C:/Users/jknau/OneDrive/Desktop/410 Applied Regression/410nba.csv", header=TRUE, sep=",")
summary(nba.data)
```

```
   PlusMinus
 Min.   :-31
 1st Qu.: -6
 Median :  0
 Mean   :  0
 3rd Qu.:  5
 Max.   : 31
```

```
 YearsLeague
 Min.   : 1.00
 1st Qu.: 2.00
 Median : 5.00
 Mean   : 5.66
 3rd Qu.: 7.00
 Max.   :16.00
```

```
      Pts
 Min.   : 0.000
 1st Qu.: 1.500
 Median : 7.000
 Mean   : 9.115
 3rd Qu.:13.000
 Max.   :52.000
```




```
      Min
 Min.   : 0.000
 1st Qu.: 5.518
 Median :21.090
 Mean   :18.859
 3rd Qu.:29.030
 Max.   :45.370
```

```
     Height
 Min.   :5.920
 1st Qu.:6.383
 Median :6.670
 Mean   :6.579
 3rd Qu.:6.830
 Max.   :7.080
```

Zero Inflated Poisson Model in R

```
nba.data$Position <- factor(nba.data$Position)
#specifying reference category
nba.rel<- relevel(nba.data$Position, ref="G")
library(pscl)
summary(fitted.model<- zeroinfl(Pts ~ Position + Height + PlusMinus + YearsLeague|Min,
                                data=nba.data))
null.model<- zeroinfl(Pts ~ 1, data=nba.data)
print(deviance<- -2*(logLik(null.model)-logLik(fitted.model)))
print(p.value<- pchisq(deviance, df=3, lower.tail=FALSE))
```

```
Pearson residuals:
    Min      1Q  Median      3Q     Max
-3.3505 -1.1712 -0.2851  0.4741 10.5056

Count model coefficients (poisson with log link):
            Estimate Std. Error z value Pr(>|z|)
(Intercept) 1.491997   0.853254   1.749   0.0804 .
PositionF   0.080807   0.074528   1.084   0.2783
PositionG   0.152739   0.098612   1.549   0.1214
Height      0.132361   0.123306   1.073   0.2831
PlusMinus   0.012162   0.002016   6.031 1.62e-09 ***
YearsLeague 0.003644   0.005970   0.610   0.5416

Zero-inflation model coefficients (binomial with logit link):
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.99186    0.58024   5.156 2.52e-07 ***
Min         -0.42872    0.07654  -5.601 2.13e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Number of iterations in BFGS optimization: 17
Log-likelihood: -760.1 on 8 Df
```

```
> print(deviance<- -2*(logLik(null.model)-logLik(fitted.model)))
'log Lik.' 213.1957 (df=2)
> print(p.value<- pchisq(deviance, df=3, lower.tail=FALSE))
'log Lik.' 5.935921e-46 (df=2)
```

The very low p-value from the deviance test confirms the model is a good fit

Model Coefficients

Zero Inflated Poisson Model in SAS

```
proc import datafile="//vdi-fileshare02/UEMprofiles/030870968/Desktop/410nbasas.csv"
     out=nbadata
     dbms=csv
     replace;
     getnames=yes;
run;
proc print data=nbadata; /* To view the imported data */
run;
proc genmod;
class Position(ref="C");
model Pts=Position PlusMinus Height YearsLeague/dist=zip;
zeromodel Min;
run;
proc genmod;
model Pts=/dist=zip;
zeromodel;
run;
data deviance_test;
deviance=-2*(-71.3892-(-57.0406));
pvalue=1-probchi(deviance,3);
run;
```

Analysis Of Maximum Likelihood Parameter Estimates

Parameter		DF	Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept		1	1.4920	0.8533	-0.1804	3.1643	3.06	0.0804
Position	F	1	0.0808	0.0745	-0.0653	0.2269	1.18	0.2783
Position	G	1	0.1527	0.0986	-0.0405	0.3460	2.40	0.1214
Position	C	0	0.0000	0.0000	0.0000	0.0000	.	.
PlusMinus		1	0.0122	0.0020	0.0082	0.0161	36.38	<.0001
Height		1	0.1324	0.1233	-0.1093	0.3740	1.15	0.2831
YearsLeague		1	0.0036	0.0060	-0.0081	0.0153	0.37	0.5416
Scale		0	1.0000	0.0000	1.0000	1.0000		

Criteria For Assessing Goodness Of Fit

Criterion	DF	Value	Value/DF
Deviance		1733.4461	
Scaled Deviance		1733.4461	
Pearson Chi-Square	198	493.3694	2.4918
Scaled Pearson X2	198	493.3694	2.4918
Log Likelihood		2617.6649	

elihood Parameter Estimates

% Confidence Limits		Wald Chi-Square	Pr > ChiSq
4517	2.5435	11371.1	<.0001
0000	1.0000		

d Zero Inflation Parameter Estimates

% Confidence Limits		Wald Chi-Square	Pr > ChiSq
4187	-0.7786	45.26	<.0001

Analysis Of Maximum Likelihood Zero Inflation Parameter Estimates

Parameter	DF	Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept	1	2.9919	0.5802	1.8546	4.1291	26.59	<.0001
Min	1	-0.4287	0.0765	-0.5787	-0.2787	31.37	<.0001

Fitted Model

π-hat = (e ^ (2.99186 - .42872 * Minutes)) /

(1 + e ^ (2.99186 - .42872 * Minutes))

And

λ-hat = e ^ (1.491997 + .080807 * Forward + .152739 * Guard +

.132361 * Height + .012162 * PlusMinus + .003644*YearsPlayed)

Analysis

- At a 5% level, minutes played significantly predicts the odds of scoring zero points. As minutes played increases by one minute, odds of scoring zero points increases by (e^(.-.42872) -1) * 100% = - 34.8657% , which is to say decreases by 34.8657%.
- At a 5% level, PlusMinus is our only variable that has a significant effect on the average points scored in a game. When PlusMinus increases by 1 point, expected points scored in the game goes up by (e^(.012162) -1) * 100% = 1.2235 %.
- ZIP was a great choice of model, but failed generate any non-obvious statistical conclusions beyond the strong but minimal relationship between PlusMinus and Pts.

Fitted Model Prediction Response

```
> random_number <- sample(1:1000, 1)
> print(random_number)
[1] 731
```

Player Performance 731 is T.J. Warren, Forward, played 9.25 minutes, height 6'8'' (6.66 feet), 9th year in the league (drafted 2014), plus minus of -5.

Prediction for points scored:
y^0 = (1 – ((e^(2.99186 + -.42872 * 9.25)) / (1 + (e^(2.99186 + -.42872 * 9.25))))) * e^ (1.491997 + .080807 + 6.66 * .132361 + .012162 * -5 + 9 * .003644) = 8.2148 points in the game

```
> print(predict(fitted.model, data.frame(Position="F", Height=6.66,YearsLeague=9,
+                                        PlusMinus=-5, Min= 9.25)))
       1
8.214772
```

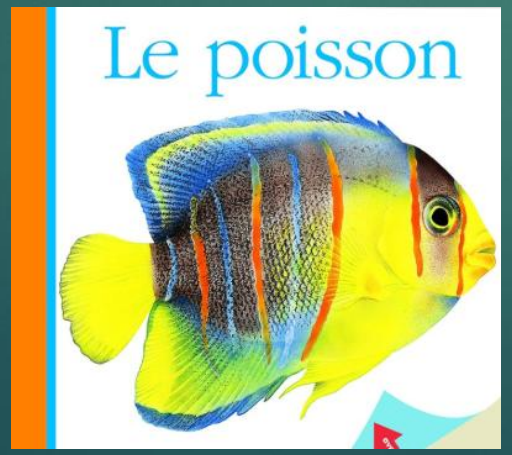
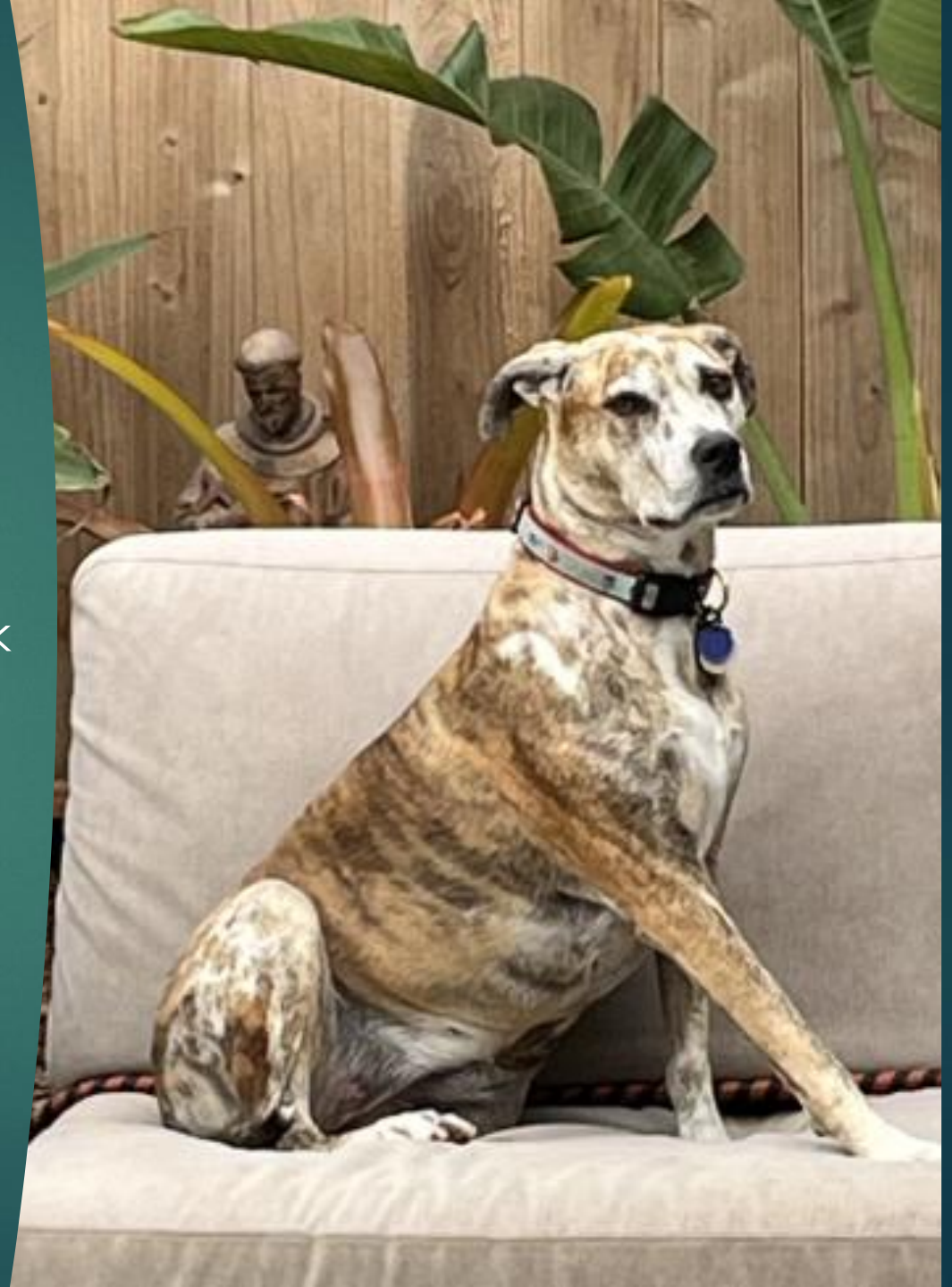
```
data prediction;
input Position$ Min PlusMinus Height YearsLeague;
cards;
F 9.25 -5 6.66 9
;
data nbadata;
set nbadata prediction;
run;
proc genmod;
class Position(ref="C");
model Pts=Position PlusMinus Height YearsLeague/dist=zip;
zeromodel Min;
output out=outdata p=pred_pts;
run;
proc print data=outdata(firstobs=202)noobs;
var pred_pts;
run;
```

T.J. Warren scored 6 points that game

The SAS System

pred_pts
8.21477

Thank you!

- Special thanks to Dr. Olga and my classmates!

Have a wonderful break everyone, good luck with the final push